Raman Biochemical Foundation Model

GAIRA V5 — Phases C1–C7 (Raman-only)

Reference corpus	375 Raman spectra · 167 pure analytes
Window / grid	450–1800 cm^{-1} @ 2 cm^{-1} (676 bins)
Representation selected	NMF (k=24) — benchmark-chosen, PCA ranked 3rd
Explained variance	0.712
Intrinsic dimensionality	~15 (90% latent var at 16)
Emergent axes	12 tentative biochemical axes
Excitation transfer	0.918 vs null 0.233
Excluded domains	Ag-SERS · Au-SERS · DART · serum Ag-colloid

KEY RESULT

A frozen, non-negative biochemical reference space built ONLY from pure Raman analytes. The same analyte measured at different laser excitations lands in nearly the same place (BSV cosine 0.918 vs 0.233 for different analytes), and unseen analytes project sensibly (BSV margin +0.62). Biological serum Raman projects onto cholesterol/albumin-like coordinates, and tube blanks show markedly less protein and fatty-acid evidence.

Unsupervised · no disease labels · no classification · nothing pushed

Executive summary

375 spectra · 167 analytes · NMF k=24

What was built

A Raman-only biochemical foundation model: 375 pure-analyte Raman spectra covering 167 analytes from three reference sources, decomposed into a frozen non-negative biochemical manifold, from which emergent biochemical axes, a Biochemical State Vector and Molecular Spectral Signatures are derived. Ag-SERS, Au-SERS and DART are excluded and remain future observation domains.

C1 — representation selected by benchmark, not by default

Five families were compared on reconstruction, stability, interpretability, replicate robustness, neighbourhood preservation and nuisance leakage, with reconstruction deliberately weighted only 10%. NMF (k=24) and ICA (k=32) tied within 0.001; the tie was broken on the pre-stated constraint that a biochemical coordinate must be non-negative and additive. PCA ranked 3rd; the autoencoder ranked 4th with component stability of only 0.16, echoing the Stage-B encoder-collapse finding; sparse dictionary learning ranked last.

C2 — the frozen manifold

Explained variance 0.712; intrinsic dimensionality about 15 effective components (90% of latent variance by 16); bootstrap component stability 0.812. The manifold is frozen and every downstream projection uses it unchanged.

C3 — emergent axes (tentative)

12 axes emerged by grouping components on the chemistry of the analytes that load on them, with literature peak assignments used only to corroborate. Coherent axes include triglycerides, saccharides, proteins, amino acids, pyrimidine bases, fatty acids, purines, polysaccharides and cofactors. Two axes remain low-confidence or unassigned and are reported as such. Themes are post-hoc interpretations, never molecular assignments.

C6 — external validation without retraining

Excitation transfer: the same analyte measured at different lasers gives BSV cosine 0.918 against a 0.233 different-analyte null (n=41). Source transfer 0.847 (n=34). Held-out analytes retain neighbourhood 0.671 and a BSV margin of +0.624. The weakest excitation transfers are ferritin, haemoglobin and albumin — heme proteins, where resonance genuinely changes the spectrum.

C7 — biological projection

477 serum Raman spectra were projected into the frozen manifold with no retraining and no labels. Their nearest pure references are cholesterol, albumin and cholesteryl esters — the expected dominant serum biochemistry. Tube blanks separate on chemistry rather than by fiat: protein evidence 0.10 versus 0.16–0.17 for serum, and fatty-acid evidence 0.00 versus 0.013–0.016. Disease groups are near-identical in these coordinates, which is reported as-is; no classification was attempted or claimed.

Scope

Raman only. The adenine concentration series, uricase and serum spike datasets named in the original plan are Ag-SERS/Au-SERS in this repository and were excluded as out-of-domain; the Raman-domain validation actually available (held-out analytes, excitation transfer, source transfer, blank control) was used instead and is documented.

Figure 1

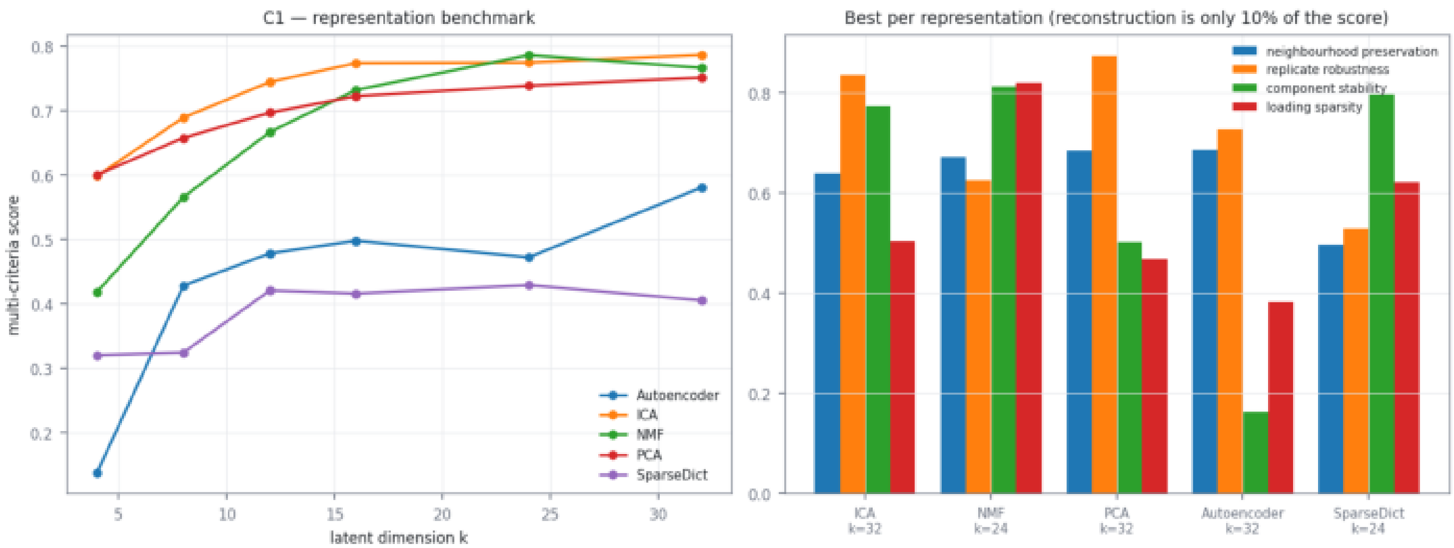

Figure 1 — C1 representation benchmark. Reconstruction is deliberately only 10% of the score; stability, interpretability and analyte structure dominate.

Figure 2

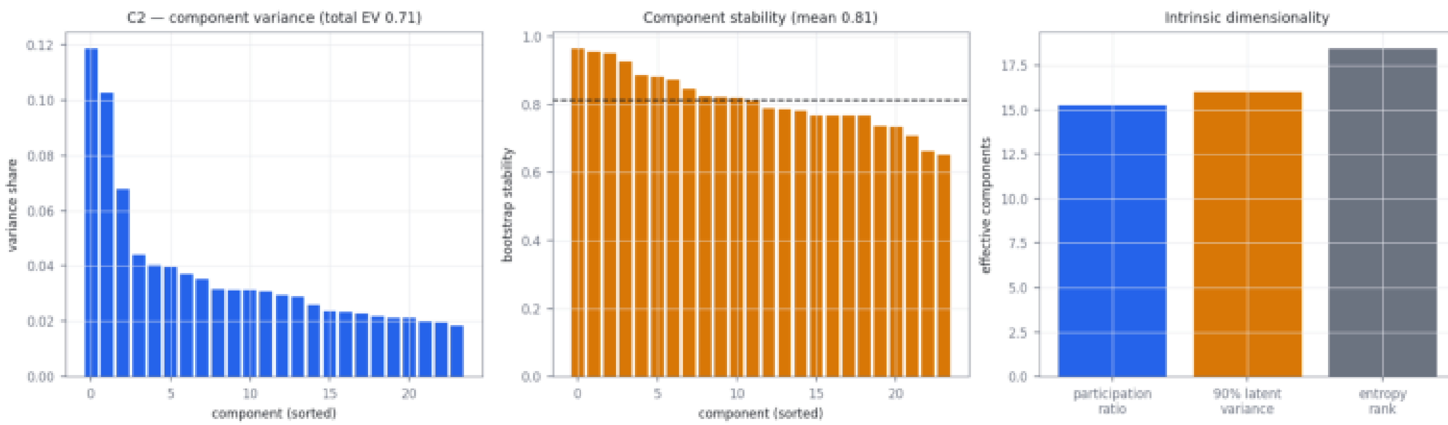

Figure 2 — C2 frozen manifold: component variance, bootstrap stability and intrinsic dimensionality.

Figure 3

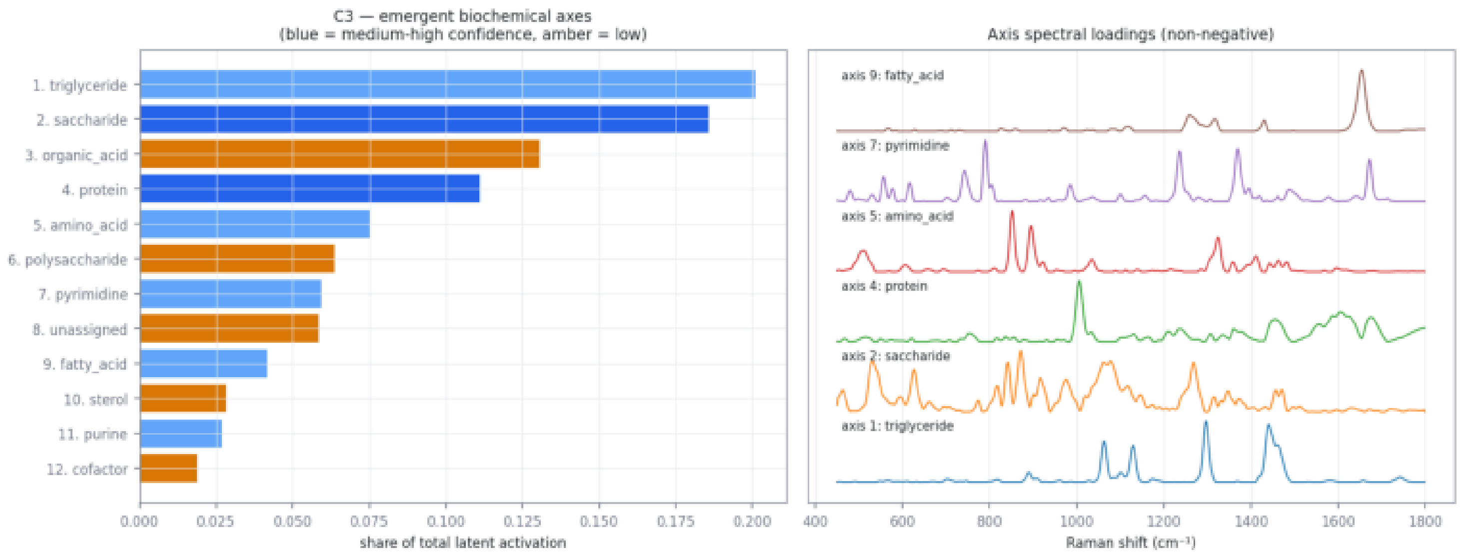

Figure 3 — C3 emergent biochemical axes and their non-negative spectral loadings.

Figure 4

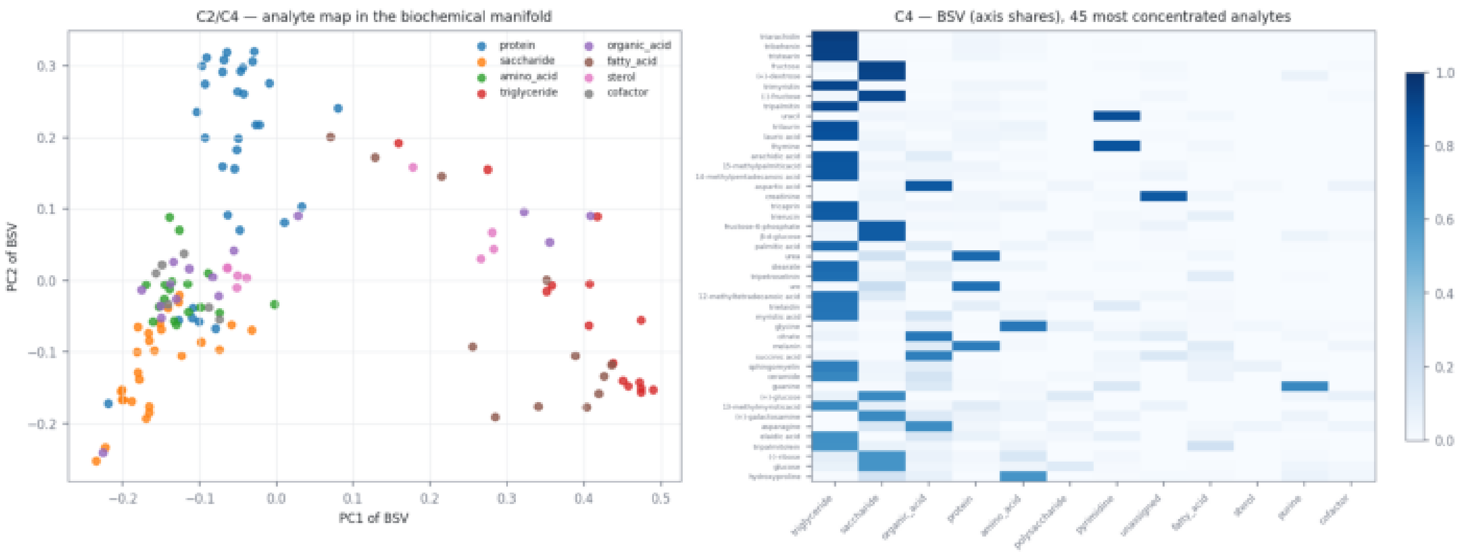

Figure 4 — Analyte map in the biochemical manifold (coloured by chemical family) and the BSV axis-share heatmap.

Figure 5

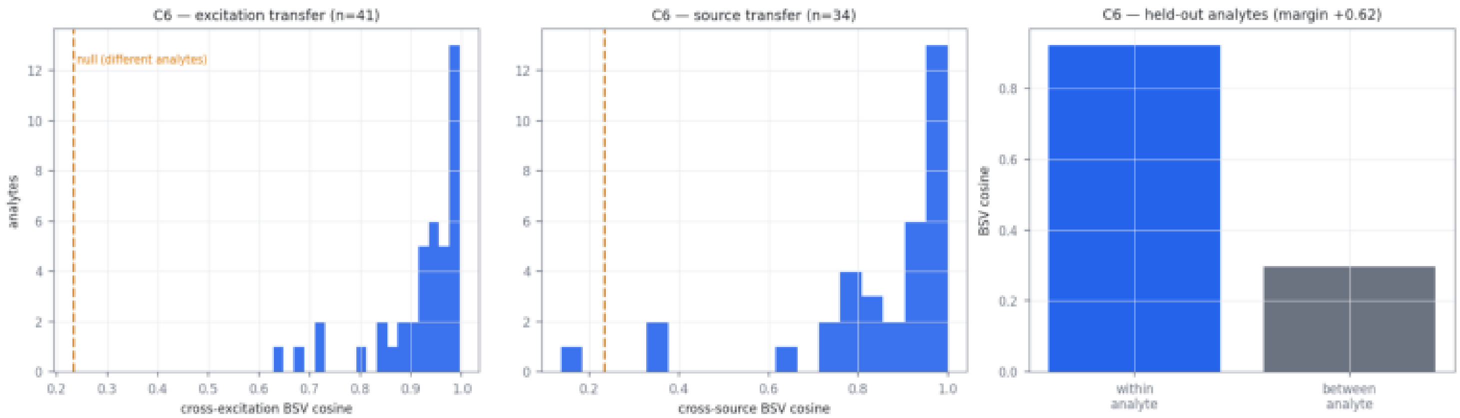

Figure 5 — C6 external validation: excitation transfer, source transfer and held-out analyte projection, each against a different-analyte null.

Figure 6

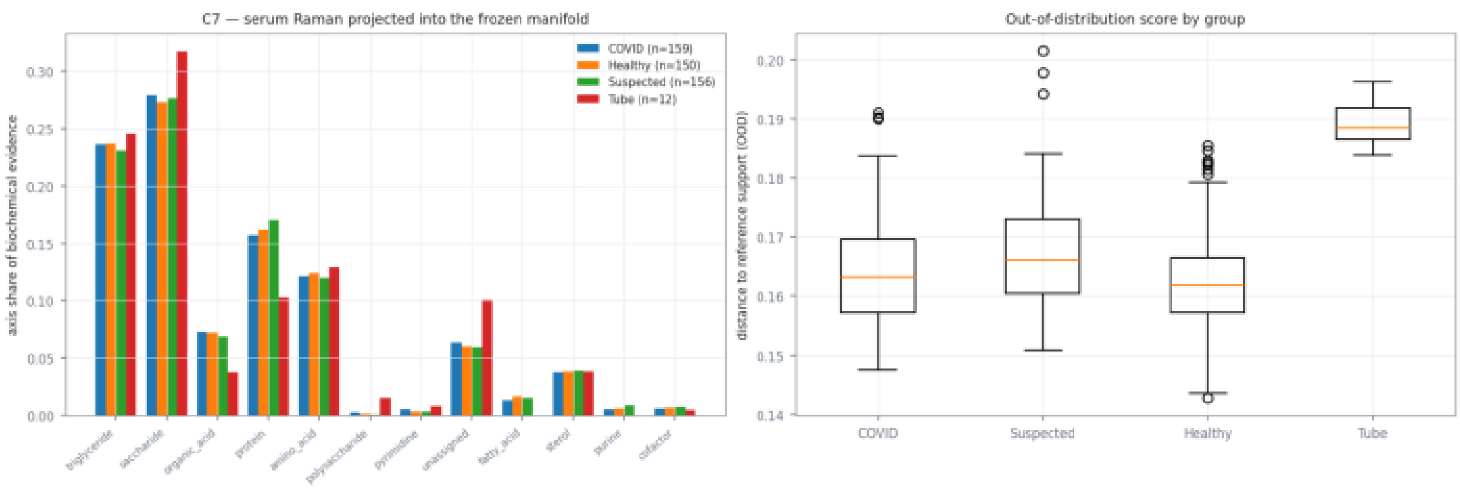

Figure 6 — C7 biological serum Raman projected into the frozen manifold, with out-of-distribution scores. Tube blanks show less protein and fatty-acid evidence than serum.

Axis catalogue and scope

Emergent biochemical axes (tentative)

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axis  1 triglyceride     n_comp= 2 share=0.201 purity=0.49 conf=medium       lit=yes
        top: tripalmitin, tristearin, triarachidin, trimyristin, tribehenin, trilaurin
        bands: 892, 1062, 1130, 1296, 1302, 1442, 1466 cm-1
axis  2 saccharide       n_comp= 5 share=0.186 purity=0.71 conf=medium-high  lit=yes
        top: fructose-6-phosphate, fructose, (-)-fructose, (+)-dextrose, β-d-glucose, arginine
        bands: 464, 468, 512, 528, 540, 576, 612, 626, 774, 818 cm-1
axis  3 organic_acid     n_comp= 3 share=0.130 purity=0.54 conf=low          lit=no
        top: citrate, aspartic acid, phosphate, succinic acid, asparagine, acetoacetate
        bands: 640, 760, 782, 866, 940, 944, 1356, 1382, 1424, 1698 cm-1
axis  4 protein          n_comp= 3 share=0.111 purity=0.63 conf=medium-high  lit=yes
        top: lectin, thaumatin, carbonic anhydrase, pepsinogen, horseradish peroxidase, superoxide dismutases
        bands: 1000, 1006, 1130, 1240, 1306, 1336, 1362, 1456, 1590, 1676 cm-1
axis  5 amino_acid       n_comp= 2 share=0.075 purity=0.46 conf=medium       lit=yes
        top: glycine, serine, hydroxyproline, alanine, glucosamine, proline
        bands: 506, 852, 896, 1034, 1322, 1326, 1360, 1412, 1444, 1464 cm-1
axis  6 polysaccharide   n_comp= 1 share=0.064 purity=0.50 conf=low          lit=no
        top: amylopectin, amylose, xylanase, lactate, lactose, cellulose
        bands: 478, 1022, 1120, 1258, 1332, 1380 cm-1
axis  7 pyrimidine       n_comp= 2 share=0.059 purity=0.43 conf=medium       lit=yes
        top: uracil, thymine, b-dna, cytosine, t-rna, a-dna
        bands: 556, 578, 616, 742, 790, 806, 984, 1236, 1370, 1396 cm-1
axis  8 unassigned       n_comp= 2 share=0.059 purity=0.28 conf=low          lit=no
        top: creatinine, tubulin, tyrosine, mannose, gluth, phosphatidylinositol
        bands: 606, 640, 674, 828, 838, 904, 1178, 1208, 1418, 1614 cm-1
axis  9 fatty_acid       n_comp= 1 share=0.042 purity=0.37 conf=medium       lit=yes
        top: arachidonic acid, trilinolenin, α-linolenic acid, linoleic acid, trilinolein, ascorbate
        bands: 1260, 1430, 1654 cm-1
axis 10 sterol           n_comp= 1 share=0.028 purity=0.32 conf=low          lit=no
        top: adenine, acetyl coenzyme a, estrone, coenzyme a, acetyl-coa, methionine
        bands: 722, 1250, 1334, 1486 cm-1
axis 11 purine           n_comp= 1 share=0.027 purity=0.56 conf=medium       lit=yes
        top: guanine, xanthine, cysteine, (+)-galactose, methionine, phosphatidylinositol
        bands: 650, 1230, 1264, 1330, 1550 cm-1
axis 12 cofactor         n_comp= 1 share=0.019 purity=0.41 conf=low          lit=no
        top: riboflavin, riboflavin, tryptophan, leucine, histidine, hemoglobin
        bands: 754, 1178, 1224, 1348, 1578 cm-1
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Next authorized action

Stop here. The Raman-only foundation model is built, frozen and validated on the available Raman-domain evidence. Ag-SERS / Au-SERS / DART integration, ontology refinement beyond the Raman corpus, biological classification and clinical prediction are explicitly out of scope and not started.